AMAZON ML CHALLENGE 2026 · TOP-1000 EVALUATION

Business Entity Resolution

Approach and methodology

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0.990654	**4.89**	**0.9975**	**$\leq$ 4B**
public leaderboard, macro $F_{0.5}$ (final)	candidates per Source-1 entity	recall of our candidate search	largest model; all MIT / Apache-2.0

Our approach in 30 seconds

1. **Clean the text** with dictionaries learned from the provided data only (Indic script, abbreviations, legal forms, house numbers).
2. **Find a few likely matches** per record: a multilingual neural search plus two exact/keyword searches, then a small model that keeps only the **2 best** candidates. This gives 4.89 candidates per S1.
3. **Score each candidate pair** with a gradient-boosted model on about 120 engineered features, refined by **four transformer "cross-encoders"** (mmBERT-small, Qwen3-1.7B twice, Qwen3-4B) that can also answer *"none of these"*.
4. **Combine and decide.** A weighted ensemble; then, for every S1, choose the match set with the highest *expected* $F_{0.5}$.
5. **Handle the unseen country** (**France**) with country-robust models, training-time augmentation, and a few rules that are each justified by training-data statistics.

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Numbers come from our run logs and saved outputs. Where a word may be new to the reader, the glossary box on the next page explains it.

Six words used throughout

S1 / Source 1	The clean reference list of businesses. We must find, for each S1, its copies in S2 and S3.
record	One row of S2 or S3 (a noisy copy of an S1 business, or an unrelated business).
distractor	A record that matches *no* S1. About 26% of training records and an estimated 40% of test records.
near-miss	The hardest distractor: a copy of an S1 with the *legal form swapped* and the *house number shifted by 1–50*, e.g. "Pacitti Medical Center | 667 Edward St" vs "Pacitti Medical Center **Co** | **674** Edward St".
candidate	An (S1, record) pair that survives the search step and is scored by the models.
$F_{0.5}$ (macro)	Computed per S1, then averaged. It weighs precision about 4× more than recall, so a wrong match costs more than a missed one. An S1 with no true match scores 1 only if we predict nothing.

1 Methodology

In short. We treat the task as *"for every record, which S1 (if any) does it belong to?"*, measure how the test data differs from the training data, build a validation set that mimics the test, and use a search → score → decide pipeline whose every design choice was chosen by measurement.

1.1 The task, as one example

Record	Address	Truth
S1: Pacitti Medical Center	667 Edward Street, Belhaven, NC	(reference)
PACITTI MEDICAL CENTER LLC	667 EDWARD STREET, BELHAVEN, NC	match (legal form added)
mpcenter.com	667 EDWARD ST, BELHAVEN, NC	match (domain name)
Pacitti Center Medical	(*no address*)	match (words reordered)
Pacitti Medical Center Co	North Carolina, Belhaven, 667 Edward St	match (address reordered)
Pacitti Medical Center Center Co	**674** EDWARD STREET, BELHAVEN, NC	**not** a match (near-miss)

Table 1: One real training entity (a validation S1) and the records that compete for it.

What makes it hard. The last record has the same name and street as the truth; only the small number shift (667 → 674) and the legal form give it away. Our final system accepts the four true records (probabilities ≥ 0.999) and rejects the near-miss (0.001).

1.2 What the data told us, and what we did about it

Finding (measured on the provided data)	Consequence for the design
Every record matches **at most one** S1, and no match crosses countries.	Solve it record by record ("which S1 owns this record?") and search within each country only.
Near-miss distractors copy an S1, swap the legal form and shift the number by 1–50. True matches change numbers by dropping or adding digits instead (87% keep the number; 55% of look-alike distractors shift it by 1–10).	Dedicated number-shift and legal-form features, and training losses that focus on the hardest wrong candidate.
The test set has **many more near-misses** than train (US 0.30 → 0.55 per S1) and more distractors overall (about 40% vs 26%).	A validation set with synthetic near-misses at test rates (F_{shift}, below), so decisions are tuned for the test, not the train.
Test adds **France**, a country with no labelled data, different formats (`N° 12`, SARL/SAS) and different vocabulary.	Country-robust models, leave-one-country-out evaluation, French-format training augmentation and train-justified rules for unseen countries.
Indic-script names are **word-by-word transliterations** (99.998% same word count).	A token dictionary learned from training pairs converts them to Latin script.
Names shared by many S1s ("chains"), and records with **no address** (4.4%).	"Chain-ness" features; the model learns not to guess when the name is not unique.

1.3 The pipeline at a glance

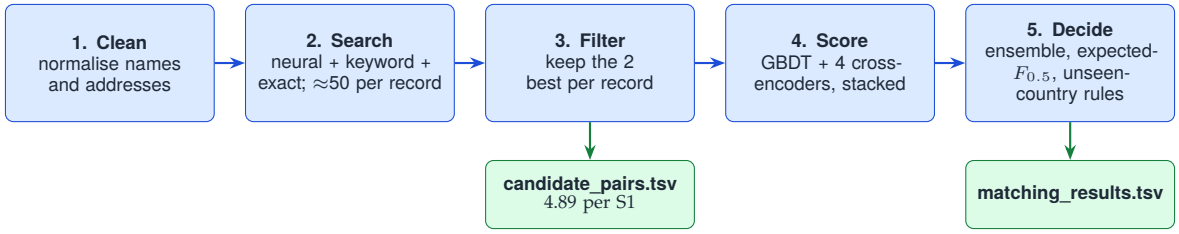


Figure 1: The five steps. Steps 1–3 are the candidate generation (Section 2); steps 4–5 are the matching model (Section 3).

1.4 How we validated without seeing test labels

We never labelled test data. Instead we built four kinds of evidence:

1. **F_{shift} — validation that looks like the test.** 20% of training S1s (441,235) are held out. We add 91,342 *synthetic near-misses* made with the generator's own operators (real number shifts and legal-form changes measured on real distractors) until each country has its test near-miss rate, and weight countries by the test mix (US 38%, India 47%, France-like 15%). Every model, threshold and weight was chosen on F_{shift}.
2. **Leave-one-country-out (LOCO).** Train on the US, test on India, and the reverse. This measures how well a model transfers to a country it has not seen, which is exactly France's situation.
3. **Labelled format checks.** The same held-out records rewritten into French formats; a robust model must score them as well as the originals.
4. **Label-free audits of every submission.** Every changed France pair is grouped into a pattern (acronym, filler word, near-miss, …) and compared with that pattern's truth rate *in training*. The public leaderboard was used only to choose between finished submissions.

2 Candidate generation / blocking

In short. For each record we search all S1s of its country with three methods (neural similarity, keyword TF-IDF and exact keys), then a small learned model keeps only the record's two most likely S1s. This keeps 98.6% of true matches while leaving just **4.89 candidates per S1** (8,478,551 pairs on test).

2.1 Step 1: clean the text

All dictionaries are **learned from the provided data** (no external resources):

- **Names:** lower-case, strip accents, undo leetspeak (harb0r → harbor), remove junk prefixes, repeated words and honorifics (Shri, Dr); split aliases ("X dba Y"); compare domain names by their stem (glyphifylaw.com → glyphifylaw); move the legal form (LLC, Pvt Ltd, SARL, ...) into its own field.
- **Indic scripts:** a 1,312-token dictionary aligned from training pairs (e.g. the Hindi spelling of "private" → private).
- **Addresses:** parse the house number (including `#`, `H.No`, `Plot No`, French `N°` and `(46)`), compound numbers (9-17-158), units and the street; canonicalise street types and states with maps learned per country (ST → street, GJ → gujarat; for France from label-free test "pseudo-pairs": R. → rue, Nord → hauts-de-france).
- Exact duplicate records (43,910 groups) are scored once and share the decision.

2.2 Step 2: search with three channels

Channel	How it works	Per record
Neural search (main)	A fine-tuned multilingual sentence encoder (`multilingual-e5-small`, 118M, MIT) turns "name | address" into a vector; we take the 50 nearest S1s of the same country by exact GPU search.	top 50
Keyword search	Word TF-IDF on the cleaned core name. Kept for records without an address.	top 20
Exact keys	Join on (house number + first street word) and on the sorted core-name words; keys shared by more than 200 S1s are skipped as uninformative.	all hits

Why a neural search. On 100k India records, recall@20 was 0.978 for the zero-shot neural encoder versus 0.72–0.90 for character TF-IDF, which also took hours instead of seconds. We then fine-tuned the encoder on training pairs (in-batch negatives from the same country, lr 10^{-5}, batch 128). It is *cross-fitted*: the model trained on half of the S1s searches for the other half, so training similarities are as honest as test ones.

Result: about 50 candidates per record and a **recall of 0.9975** on validation (0.9996 when the record has an address, 0.952 when it does not).

2.3 Step 3: a learned filter produces `candidate_pairs.tsv`

A GPU XGBoost model (25 cheap features: search scores and ranks, name/street similarity, house-number equality and difference, legal-form relation) gives each candidate a probability p_1. Each record keeps its **top 2 candidates with** $p_1 \geq 0.01$. After the organisers announced that smaller candidate sets rank higher, we chose this rule from the measured trade-off, *retraining the matcher on each option*:

Keep rule	per test S1	recall ceiling	matcher F_{shift}
top-12 or $p_1 \geq 0.001$ (first version)	69.3	0.9945	0.9855
top-16 and $p_1 \geq 0.001$	7.88	0.9943	0.9871
top-2 and $p_1 \geq 0.01$ (final)	**4.89**	0.9860	**0.9872**

2.4 How we made sure true matches were not lost

- Each channel was added only if it raised measured recall; the union reaches 0.9975.
- The keep rule was chosen by the *final* score after retraining, not by recall alone: going from 69.3 to 4.89 candidates *did not lower* F_{shift} (shorter lists make the list-level features sharper).
- The true pairs the tighter rule drops are ones the matcher never selected anyway: almost all are records without an address whose name is shared by several S1s.
- `candidate_pairs.tsv` is written from the very same object the models score, so it is exactly the set used at inference.

A France test record. "Cussac SARL & Associés | 48 Rue Gustave Deloy, Nord, Tourcoing" gets 69 candidates from the search. The true S1 "Cussac Btp SARL | 48 Rue Gustave Delory, Tourcoing" ranks first (similarity 0.80; the next is 0.52), and the filter keeps only it ($p_1 = 0.91$).

3 Model architecture and feature engineering

In short. A gradient-boosted model scores every candidate pair from about 120 engineered features. Four transformer cross-encoders read the two texts side by side and add their judgement. A second gradient-boosted model combines everything ("stacking"). Four such stacks are averaged, and a per-S1 decision picks the match set with the highest expected $F_{0.5}$.

3.1 Architecture

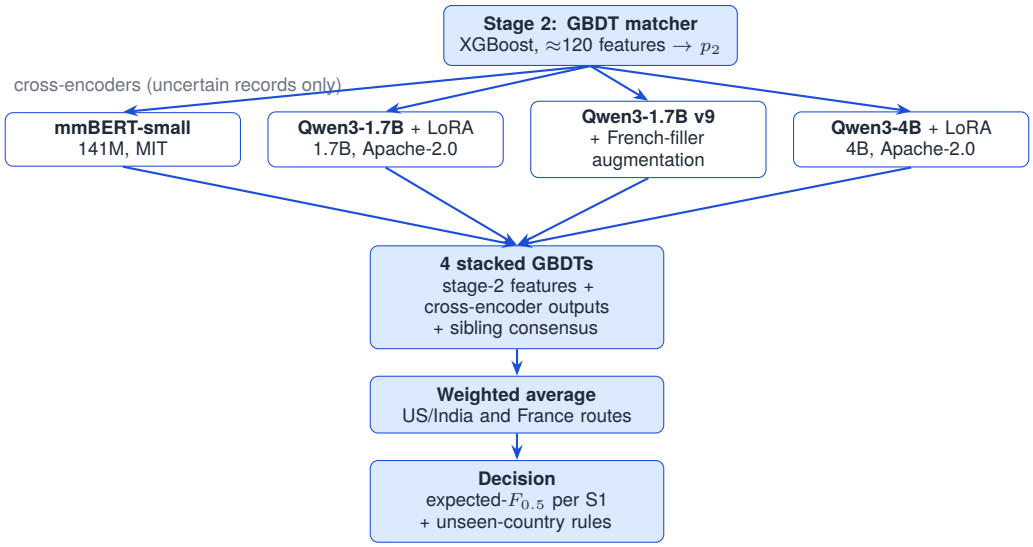


Figure 2: The matching model. Each cross-encoder has its own stacked model; the four stacked probabilities are averaged.

3.2 Feature engineering

Group	Features	What it catches
Name similarity	ratio, partial, token-sort, token-set, Jaro–Winkler, Levenshtein; no-space and domain-stem similarity; alias match; first-word match	typos, reordering, domains, aliases
Acronym	record name equals the S1's initials	"AL" for "Alternative Loisirs SA"
Near-miss block	house number equal; signed and absolute difference; relative difference; *edit type* (digit dropped / inserted / substituted / small arithmetic shift); compound-number tuple; bis/ter; units; legal-form relation	667 vs 674 with LLC vs Co (a distractor) vs 3391 vs 391 (a real typo)
Address	street and address similarity, component overlap, state included, address missing	reordered or partial addresses
Chain-ness	how many S1s share the S1's / record's name	whether a name-only record can be trusted
Open-set scores	tail statistics of the record's candidate scores (extreme-value / libMR), top score and margin, Extreme Value Machine inclusion probability, ranks and gaps	"is the best candidate a clear winner?"
Vocabulary (label-free)	*filler* words mined per country (US/India: center, partners, services; France: groupe, développement, associés, et fils); *real-word* test for the record's extra words	filler = same business; real word swapped = a different, sibling business
Stacked only	cross-encoder outputs; agreement of models on the best S1; *sibling consensus* (other records confidently assigned to the same S1 and whether they share this record's name or number)	groups of records confirming each other

The features at work (real values). For the true record "PACITTI MEDICAL CENTER | 667" and the near-miss "Pacitti Medical Center Center Co | 674", the name and street similarities are *identical* (100). The number features differ: shift 0 vs +7, edit type "equal" vs "small arithmetic shift". The model gives 0.99999 vs 0.0025.

3.3 The models

Stage 2 — GBDT matcher. GPU XGBoost with the configuration of the 4th-place Foursquare Location Matching solution (lr 0.05, depth 15, column sample 0.5), 5-fold cross-validation grouped by S1 so that every training prediction is out-of-fold.

Cross-encoders. Each reads a record together with its whole candidate list plus a learned **"none of these" (NIL)** option, e.g. for Qwen:

```
Business A: Cussac SARL & Associés | 48 Rue Gustave Deloy …
Business B: Cussac Btp SARL | 48 Rue Gustave Delory …
Same business:
```

The training loss has four parts, each targeting a known failure:

- **list loss** — pick the right candidate or NIL (teaches "none of these");
- **one-vs-all loss on the hardest wrong candidate** — usually the near-miss copy (from OpenMatch, NeurIPS 2021);
- **max-wrong-score penalty** (weight 4) — an error happens when the *best* wrong candidate is too high (from Open-Set Biometrics, ECCV 2024);
- **auxiliary heads** (weight 0.5) predicting number, legal-form and street equality, so the model looks at those fields.

Only "uncertain" records (stage-2 best probability between 0.01 and 0.99) are sent to the cross-encoders.

Cross-encoder	Training	Dev AP (2 folds)
mmBERT-small (141M, MIT)	full fine-tune, lr $3 \cdot 10^{-5}$; chosen from a bake-off of 10 licence-compliant models	0.9977 / 0.9982
Qwen3-1.7B + LoRA	LoRA rank 64, lr $2 \cdot 10^{-4}$, bf16; $\leq$300k records per fold	0.9984 / 0.9977
Qwen3-1.7B v9	continued with **French-format and French-filler augmentation** (below)	0.9983 / 0.9981
Qwen3-4B + LoRA	2 folds × **352,000 records** (11,000 steps of 32), 2 GPUs per fold, French-format augmentation	**0.9985 / 0.9977**

Why augmentation. France has no labels, and its formats never occur in training. During training we randomly rewrite a record *and all its candidates* with one consistent mapping into French forms, keeping the label: "Harbor Brokerage Inc | 1204 Lake Street" → "Harbor Brokerage SARL | N° 1204 Lake Street"; "Balaji Services" → "Balaji Groupe". On held-out records rewritten into French formats, the accuracy loss fell from 0.008 to 0.0006 (dev AP).

Stacking and ensemble. For each cross-encoder, a second XGBoost learns from the stage-2 features plus that model's outputs and the sibling-consensus features. The four stacked probabilities are averaged with weights chosen on validation, after checking all 31 subsets (every model helped):

Route	mmBERT-small	Qwen3-1.7B	Qwen3-4B	v9	F_{shift}
US and India	1.5	1.5	3	0.75	0.99179
France (unseen)	1.5	2	3	1	0.99127

Stacking fixes what single models miss (real values). A near-miss "Sales Csnuafltnams PVT LTD | ##128" next to the true "Sales Consultants Private Limited | 137" gets 0.21 from stage 2; after the cross-encoders and stacking it drops to 0.0007 and is rejected.

3.4 The decision

Each record goes to its single best S1. For each S1 we then compute, exactly, the **expected $F_{0.5}$** of predicting its top $1, 2, \ldots$ records, and choose the best:

Probabilities of an S1's records	expected $F_{0.5}$ for 0, 1, 2, 3 records	decision
0.9, 0.6, 0.3	0.03, **0.77**, 0.75, 0.64	predict only the first
0.7, 0.7	0.09, 0.62, **0.72**	predict both
0.2	**0.80**, 0.20	predict nothing

3.5 Country routing and unseen-country rules

Two model versions are trained: *full* (all features) for countries seen in training, and *transfer* (without features whose values are new in another country, chosen by leave-one-country-out) for France. Four write-time rules apply to unseen countries. Each is **re-derived from training labels every time** and is only used when the training evidence is clear:

Rule	Pattern	Truth in training	Effect on France (final)
Sibling guard	a real word swapped at the same address ("Dj Collège" vs "Dj Club")	US 77%, India 8% (inconsistent, below the 80% break-even)	6,595 pairs not matched
Near-miss guard	same name, different legal forms, number shifted 1–50	$\leq 1\%$ (India 0.0%, US 0.9%)	23 pairs not matched
Country-word guard	S1 name + the country's name ("Ets Lhippocampe SAS [France]")	0.2%	100 pairs not matched
Filler rule (adds)	category word replaced by the country's filler words, same address ("Cussac Btp SARL" → "Cussac SARL & Associés")	US 99.5%, India 97.8%	4,053 pairs added

4 Other relevant information

4.1 Results over the challenge

Submission	Main change	Public LB
v1	first full pipeline (7.88 candidates per S1)	0.98030
v4	2-candidate rule; country routing; filler and real-word features	0.98830
v5	larger mmBERT-base + self-training on France (a regression)	0.98791
E2h	+ Qwen3-1.7B ensemble + unseen-country guards	0.99001
E5a	+ French house-number parser; French-format augmentation	0.99008
G3	+ Qwen3-4B, filler augmentation, filler rule	0.99054
G5 (final)	**4-model weighted ensemble**	**0.990654**

The largest gains (v1 → v4 → E2h) came from diagnosing and fixing the unseen country, not from bigger models.

4.2 Error analysis

- **Missed matches:** about 80% are records *without an address* whose name is shared by several S1s (e.g. "Smith (Surgical)" vs three "Smith Surgical LLC" in different states). Under $F_{0.5}$ the model correctly declines to guess.
- **Wrong matches:** residual near-misses, name-only records of chains, and different businesses in the same building.
- **France** remains the least certain country (estimated $F_{0.5} \approx 0.981$ vs ≈ 0.996 for the US and India).

4.3 What we tried and dropped

Character n-gram blocking (too slow, lower recall); CatBoost blend (weight 0); isotonic calibration (not better); self-training on France (lost on the leaderboard); a France-specific prior shift (added risky guesses).

4.4 Compliance

- **Models:** `multilingual-e5-small` (MIT, 118M), `mmBERT-small` (MIT, 141M), `Qwen3-1.7B` and `Qwen3-4B` (Apache-2.0, $\leq$ 8B); XGBoost and LightGBM (Apache-2.0 / MIT). Revisions are pinned.
- **No external data or APIs.** Every dictionary is learned from the provided files. Unlabelled test data is used only for vocabularies (e.g. French street abbreviations), never with labels. *Disclosed:* the French-format augmentation uses a short hand-written list of French surface forms (legal forms, `N°`, the word "France", filler words).
- **No test labels.** Test records were read only to debug; every rule derived from that reading was confirmed on training labels. The leaderboard was used only to choose between finished submissions.

4.5 Reproducibility

- Code: `code/business_entity_resolution/` (24 modules in `src/`, 20 unit tests).
- `./run_all.sh` runs every stage from the raw data; `./run_final_G5.sh` trains the other cross-encoder runs and builds the final ensemble. Both are resumable by stage.
- From the saved runs, `START_FROM=ensemble ./run_final_G5.sh` reproduces the submitted files byte for byte; the official validator passes.
- The Qwen3-4B training code, launch scripts, manifest and logs are in `qwen4b_remote_training/`.

4.6 Research that shaped the design

Source	What we took
Foursquare Location Matching, 4th place (Kaggle 2022)	search $\rightarrow$ filter $\rightarrow$ GBDT cascade and its GBDT settings
BLINK (EMNLP 2020); Ditto (VLDB 2021)	bi-encoder search + cross-encoder re-ranking; training settings
NIL entity linking (Sil & Florian 2016; Zhu et al. 2023)	the "none of these" option and auxiliary heads
OpenMatch (NeurIPS 2021); Open-Set Biometrics (ECCV 2024)	hardest-negative and max-wrong-score losses
Meta-recognition, Extreme Value Machine (TPAMI 2011/2018)	open-set score features
QLoRA (Dettmers et al. 2023); Jansche (2007)	LoRA settings; optimal expected-F decision